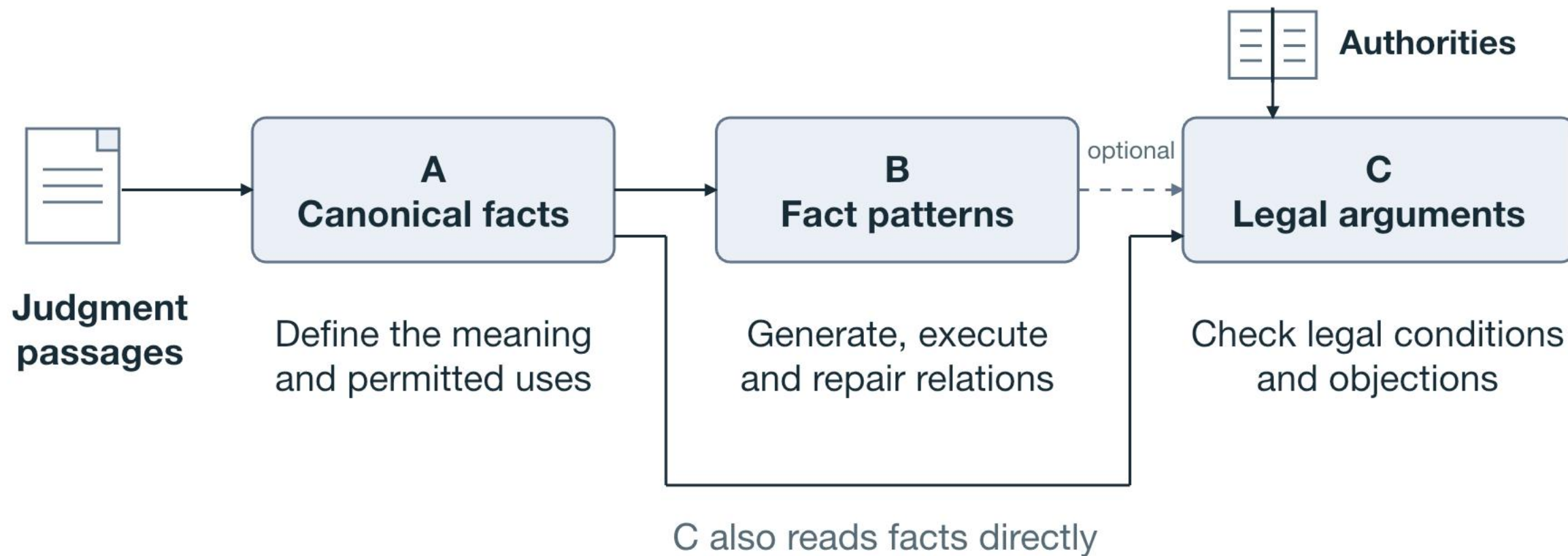


Legal fact algorithms

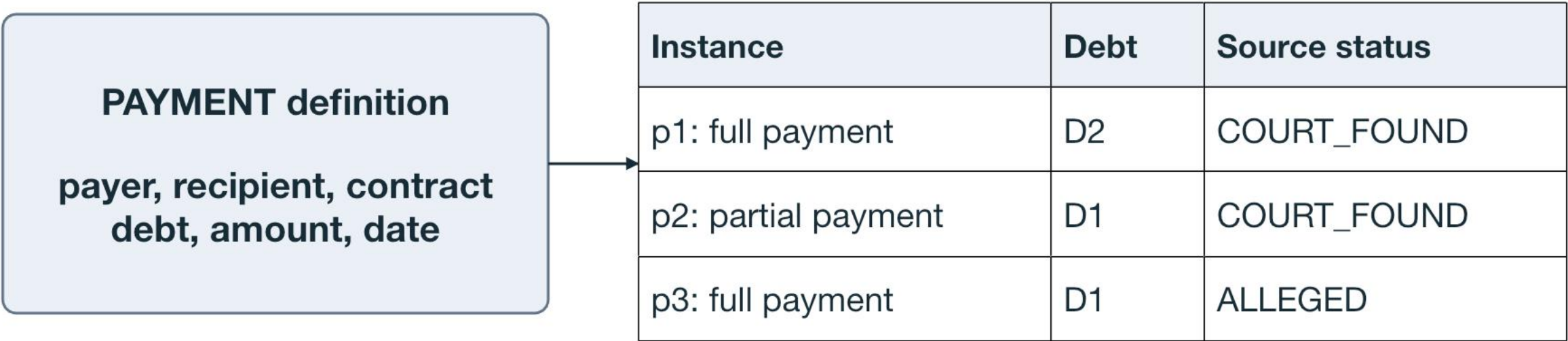
A research design for comparing cases and reconstructing legal arguments



A. A shared definition with preserved differences

A common fact type keeps each instance's identity, parameters and source status

Source assertion → Retrieve definitions → Check each slot → Map or keep pending



Same type Different parameters and assertions remain distinct

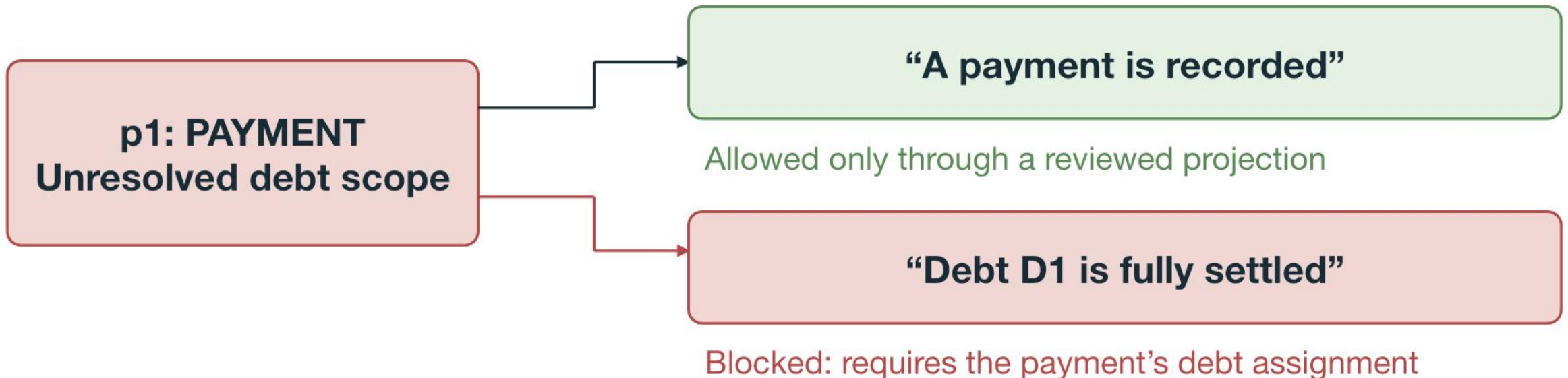
A. A qualifier blocks only the uses it affects

Each query declares which fields and semantic conditions it requires



“Full payment was made, but only for debt D2.”

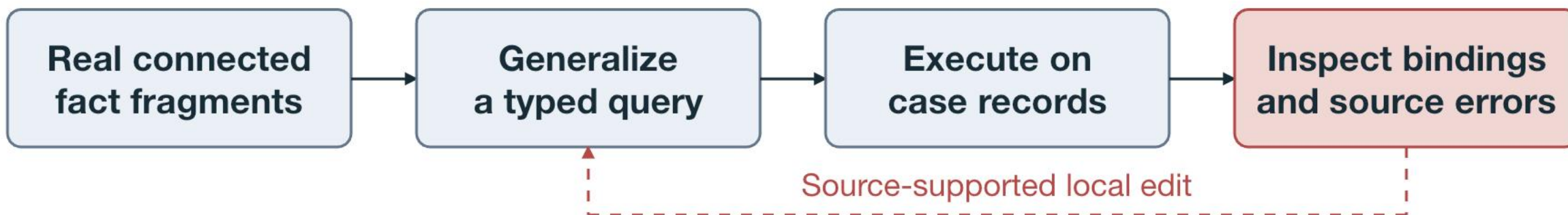
The debt qualifier is still unparsed.



After source review: p1 binds to D2. Another complete D1 payment remains usable. Green: allowed. Coral: blocked.

B. Relation discovery with execution feedback

Keep the intended meaning fixed while repairing the query



Wrong binding

notice: K1 / D1
payment: K1 / D2

Shared contract AND debt

ReceivedNotice(n, K, D, t1)
FullSettlement(p, K, D, t2)



B. Reusable relations and a compact dictionary

A relation can remain useful even when it saves no space



Define once

P = receipt + full payment
for the same contract and debt

Pay for every use

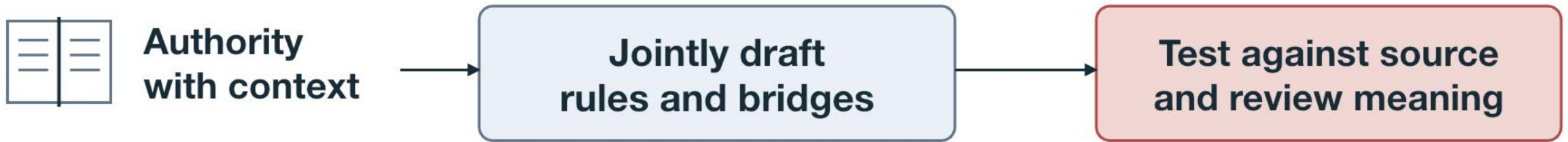
P(K1, D1, n1, p5, dates...)
+ source, status and residual fields

Minimize definition bits + use bits + remaining record bits

Each base record reconstructs once. Shared entity IDs are allowed. Rare legal conditions remain available to C.

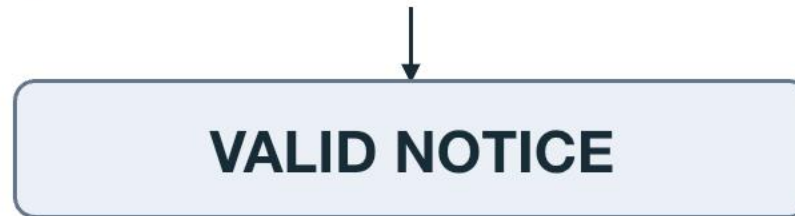
C. Legal rules require sourced bridges

A descriptive event does not automatically satisfy a legal condition



Bridge

Actual receipt + designated recipient + clear debt identity



Rule

Valid notice + full payment within the defined period
supports a specific protection, subject to exceptions

C. One witness, two exception policies

Hypothetical rule: protection for full payment within 10 days after valid receipt

K1 / D1

Sent May 1

Received May 4

Paid May 13

—————→
9 days

Reviewed policy	Revocation unknown	Result for this route
Exception needs supporting evidence	No qualified exception argument available	Accepted under default*
Explicit absence is required	No evidence that payment was not revoked	Missing premise

* Report the default and the unknown exception together.

All other required premises hold in this example. The result concerns one termination ground, not the whole case.

First prototype and research hypothesis

Can use-specific checks preserve valid evidence while reducing semantic errors?

First implementation

One issue and procedural stage
Shared facts, definitions and identities
Executable relations and a small rule package

Critical comparison

Ignore the qualifier
Block the whole assertion
Block only affected uses

Evaluate together

Incorrect uses, valid witnesses retained,
unresolved cases and review time